Electronic correlations on the verge of the Mott transition in $La_{1-x}Sr_xTiO_3$ by $^{47/49}Ti$ and ^{139}La nuclear magnetic resonance

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Carrier-doping effects on electronic correlations on the verge of the Mott transition in $La_{1-x}Sr_xTiO_3$ have been investigated by $^{47/49}Ti$ and ^{139}La NMR. In antiferromagnetic (AF) insulator of $LaTiO_3$ with $T_N = 140$ K, we observed zero-field Ti NMR and determine the hyperfine field of 91.4 kOe for Ti nuclei. The increases of nuclear spin–lattice relaxation rates $(T_1)^{-1}$ of La and Ti show that effective carrier mass is enhanced critically as the Mott transition boundary is approached from the metallic side. We have no evidence for a prominent increase of AF spin fluctuations at $q \sim Q$ near the Mott transition boundary. Together with the temperature dependence of T_1 (The Korringa relations of $T_1T =$ const are observed for La and Ti) in the metallic state, we conclude that the AF spin fluctuations of Ti-3*d* spins due to the interatomic electronic correlations is not important, but single-site (intraband) electronic correlations play an important role in driving the metal-insulator transition in $La_{1-x}Sr_xTiO_3$, as opposed to the case of high-T_c cuprates. [S0163-1829(99)13415-5]

I. INTRODUCTION

Extensive studies to understand the mechanism of high-T_c superconductivity (HTSC) with layered perovskite structure have led to renewed interest in the narrow band systems of 3*d* transition-metal (TM) oxide.[1–5] The one-site Couloumb interaction *U* and the one-electron band width *W* control physical properties of the system. According to the Hubbard model,[6] the system undergoes a phase transition between a correlated insulator (Mott insulator[7]) and a correlated metal at a critical value of *U*. Brinkmann and Rice[8] described metallic state as a strong renormalized Fermi liquid and showed that at half filling, $n=1$, intraatomic Couloumb interactions cause an divergent enhancement of the electron mass. Along the attempt of closing band gap, pressure effect, or alloying effects on a number of oxides such as V_2O_3 (Refs. 3 and 9) have been investigated so far. The band filling through carrier doping also convert the insulating to the conducting state. The carrier doping for the 3*d* oxides as well as the high-T_c cuprates has been investigated intensively to understand systematics of the change of electronic state. Especially, a lot of studies on the titanium oxides, one of the most appropriate insulator-metal transition systems, has been performed to elucidate the band-filling or the gap-controlling effects on electronic and magnetic properties.[5,10–22]

According to the picture for a general framework for oxides by Zaanen, Sawatzky, and Allen,[23] there are two types of band gaps possible. The parent cuprate oxides as well as the most late-3*d* TM oxides are classified as the charge-transfer-type insulators where the lowest charge excitations are from oxygen 2*p* to TM-3*d* states. On the other hand, some light-TM oxides are known as Mott-Hubbard insulator in which a possible charge gap is formed between the Coulomb split-*d* bands. Photoemission studies showed Mott-Hubbard gap is 0.1 eV for $LaTiO_3$, and the band gap changes systematically from Ti to Ni.[24]

The compound $LaTiO_3$ with orthorhombically distorted perovskite structure (so-called $GdFeO_3$-type structure) is known to be a typical Mott-Hubbard insulator with antiferromagnetically ordered state below the Neel temperature T_N of 120–140 K.[10–14] The valence number of Ti ions is $3+(d^1)$ and one *d* electron is localized due to strong electron correlations. When one changes the formal valence of titanium from $3+(3d^1)$ to $4+(3d^0)$ by alloying the trivalent La (Y) ions and divalent Sr (Ca) ions in the system $La_{1-x}Sr_xTiO_3$ or $Y_{1-x}Ca_xTiO_3$ undergoes the Mott insulator-metal transition.[5,15–19,22] Off-stoichiometry of oxygen or La in $LaTiO_3$ also causes the metal-insulator transition.[13,14,20]

Measurements of magnetic susceptibility,[5,16,17] electronic specific heat,[5,18,19] Hall coefficient,[5,16] and optical conductivity[16,22] for $La_{1-x}Sr_xTiO_3$ and $Y_{1-x}Ca_xTiO_3$ indicate that one of characteristic properties in the metallic states is the divergent enhancements of effective carrier mass near the phase boundary of metal-insulator transition. The Wilson ratio (χ/γ in a unit of $3\mu_B^2/\pi^2k_B^2$) is nearly constant (~ 2) against *x*,[5,16,17] indicative of absence of the Stoner enhancement even when the system approaches the Mott insulator boundary. These results are pointed out to be sharply contrastive to the system with predominant paramagnon interactions and also the cuprate oxides.[19,25]

NMR is a powerful tool to extract information of mag-

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netic fluctuations at the individual atomic sites in the system. In the case of HTSC, antiferromagnetic (AF) spin correlations are inferred from the large enhancements of nuclear spin-lattice relaxation rate T_1^{-1} observed at Cu sites.[26–28] In this study, we have carried out NMR of $^{47/49}$Ti and ^{139}La in $La_{1-x}Sr_xTiO_3$ to investigate microscopically effects of carrier doping on q wave-vector-dependent electronic correlations. We focus our attention to whether AF spin fluctuations of Ti^{3+} spins in the metallic state near the Mott-transition exist or not, and to compare with the case of HTSC with the charge-transfer type gap.

II. EXPERIMENT

$La_{1-x}Sr_xTiO_3$ compounds were melt grown by a floating-zone method in a reducing atmosphere. Starting materials of Ti, TiO_2, La_2O_3, and $SrTiO_3$ were weighed for appropriate amounts, mixed thoroughly and pressed into rod. The bar shaped ingredient were melted under Ar/H_2 atmosphere in a floating-zone furnace. Detail conditions of the heating were reported elsewhere.[16,20] A x-ray diffraction measurement indicates that the samples were of single phase, and that the diffraction peaks were indexed with orthorhombic or cubic-lattice structure. The cubic to orthorhombic structural change was observed to take place around $x=0.4$ at room temperature. Upon the structural change, however, we have not observed distinct change in lattice constants and electronic properties (electrical resistivity, magnetic susceptibility, and specific heat[5,16,19]). $LaTiO_3$ is confirmed to be an antiferromagnetic insulator with $T_N \sim 140$ K by magnetic susceptibility and specific heat measurements.[16,19] For the Sr-substituted samples, the magnetic susceptibility is nearly temperature independent and is considered to be of the typical Pauli paramagnetism.[5,16,19] The temperature dependence of electrical resistivity shows that metal-insulator transition occurs near $x \sim 0.05$.[5,15,16]

NMR was carried out by utilizing a phase-coherent spin-echo pulse spectrometer and a superconducting magnet. For a part of Ti NMR, a spectrometer with the superconducting magnet of 15 T in Tohoku University was used in order to improve sensitivity. Properties of ^{139}La nuclei are nuclear spin of $I=\frac{7}{2}$, nuclear quarupole moment of $Q=0.21(10^{-24}\,\mathrm{cm}^2)$ and gyromagnetic ratio of $\gamma_N/2\pi=6.0144\,(\mathrm{MHz/T})$. For ^{47}Ti and ^{49}Ti, $I=\frac{5}{2}$ and $\frac{7}{2}$, $Q=0.29$ and $0.24\,(10^{-24}\,\mathrm{cm}^2)$ and $\gamma_N/2\pi=2.4000$ and 2.4005 MHz/T, respectively.[29]

III. RESULTS

A. NMR of antiferromagnetic insulator $LaTiO_3$

Figure 1 shows $^{47/49}$Ti and ^{139}La spectrum of $LaTiO_3$ in antiferromagnetic-ordered state under zero external field. The two resonance peaks near 7 and 11 MHz with relatively strong intensity shown in Fig. 1(a) are assigned to ^{139}La NMR and other signals above ~11 MHz with poor intensity (almost 10 times smaller) shown in Fig. 1(b) are attributed to $^{47/49}$Ti, NMR as discussed below.

The NMR spectra for $^{47/49}$Ti can be analyzed for the case that the Zeeman interaction, H_M, is much larger than the quadrupole interaction, H_Q, i.e., $H_M \gg H_Q$. The Hamiltonian is expressed as follows,[29]

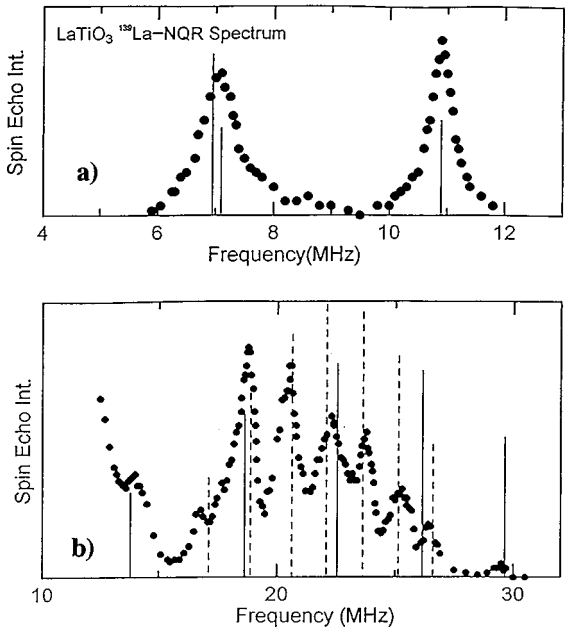


FIG. 1. NMR spectrum under zero external field at 1.4 K for the antiferromagnetic state of $LaTiO_3$. (a): for ^{139}La NMR, and (b): for $^{47/49}$Ti NMR. The solid and broken lines indicates the calculated resonance positions (see text).

$$H = H_M + H_Q, \tag{1}$$

and

$$H_M = -\gamma_N h \mathbf{H}_{\mathrm{int}} \cdot \mathbf{I}, \tag{2}$$

$$H_Q = (1/6) h \nu_Q [3I_Z^2 - I(I+1) + (1/2)\eta(I_+^2 + I_-^2)], \tag{3}$$

where $\nu_Q = (3/2)e^2qQ/I(2I-1)$, γ_N is gyromagnetic ratio of nuclear moment, and η is asymmetric parameter of electric-field gradient (EFG) defined by $(V_{XX}-V_{YY})/V_{ZZ}$, $|V_{ZZ}| \geqq |V_{YY}| \geqq |V_{XX}|$. H_{int} is an internal magnetic field at a polar angle θ with respect to the principal axis of V_{ZZ}. The maximum principal axis at the Ti and the La site in $LaTiO_3$ is determined to be the b axis from a calculation of EFG by a point charge model.[30]

With the second-order perturbation calculation or a numerical calculation for Eq. (1), the resonance positions for the each transition for the Ti nuclear levels can be fitted to the peaks above 12 MHz. The following set of parameters are obtained: $H_{int}=91.4$ kOe, $\nu_Q=5.9$ MHz for ^{47}Ti, and 2.3 MHz for ^{49}Ti, $\eta \sim 0.35$, and $\theta \sim 90°$. Solid and broken lines in Fig. 1(b) are corresponding to the calculated resonance peak positions for ^{47}Ti and ^{49}Ti sites, respectively. The fitting between experimental data and calculated resonance peaks seems to be quite well as seen in Fig. 1(b). The internal field is given as $H_{int}=A_{hf}\langle S\rangle$, where A_{hf} is a hyperfine coupling constant and $\langle S\rangle$ is an Ti^{3+} ordered moment. Assuming that a main source for the hyperfine coupling is attributed to a core polarization, of which the hyperfine-coupling constant is $A_{hf}=100 \sim 120\,\mathrm{kOe}/\mu_B$,[31] the ordered moment is estimated to be $0.9 \sim 0.8\mu_B$. This value is close to the one ($1\mu_B$) expected for $S=\frac{1}{2}$ of Ti^{3+}.

For ^{139}La ($I=7/2$), one might expect three nuclear quadrupole resonance (NQR) lines (in the case of $H_M \ll H_Q$) or seven quadrupole split Zeeman resonance lines (for H_M

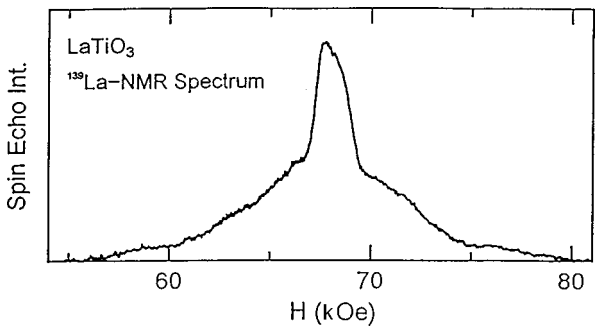


FIG. 2. ^{139}La-NMR spectrum as a function of magnetic field for the antiferromagnetic state of $LaTiO_3$ obtained at $T=40$ K and at $f=40.64$ MHz.

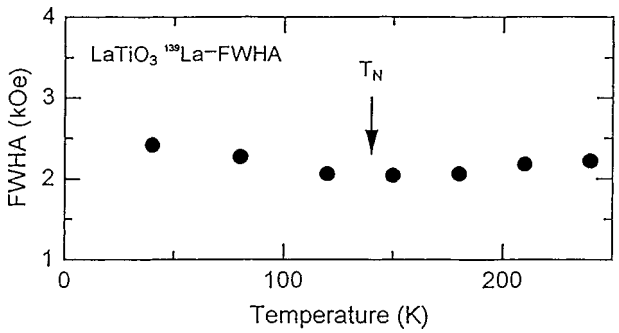


FIG. 3. Full width at half amplitude of the central resonance line of ^{139}La NMR for $LaTiO_3$ as a function of temperature.

$\ll H_Q$). Beside with the Ti signals discussed above, we got only two peaks at near 11 and 7 MHz as shown in Fig. 1(a). This can be interpreted only in the case of $H_M \ll H_Q$ with large η. From fitting the spectrum by the Hamiltonian of Eq. (1), we obtain the parameters of $\nu_Q=3.8$ MHz and $\eta=0.6$ and nearly vanishing value of H_{int}. In this case, two resonance lines are almost superposed at lower peak as indicated by the solid lines in Fig. 1(a). Here, we do not need any finite value of magnetic field at the La site for fitting, indicating that the internal field from the antiferromagnetically ordered Ti moments is completely canceled out at the La site. This may be originated from the magnetic and structural features in $LaTiO_3$.

Figure 2 shows ^{139}La-NMR spectrum under an external magnetic field at 10 K and the resonance frequency $\nu_0=40.64$ MHz. As the NMR measurement was carried out for powder samples that contain small grains with randomly oriented crystalline axis, the NMR spectrum is obtained as a so-called powder pattern. In this case, the second-order perturbation calculation of Eq. (1) shows that the NMR line of the central transition ($|1/2\rangle \leftrightarrow |-1/2\rangle$ splits into two singular positions and the separation of the two singularities, ΔH is given by[29]

$$\Delta H=\nu_Q^2[I(I+1)-3/4](\eta^2+22\eta+25)/144\nu_0. \quad (4)$$

Actual central line of the spectrum with those singularities is broadened inhomogenously. Satellite contributions ($\pm 1/2\rangle \leftrightarrow \pm 3/2\rangle$, $\pm 3/2\rangle \leftrightarrow \pm 5/2\rangle$, $\pm 5/2\rangle \leftrightarrow \pm 7/2\rangle$) spread over 20 kOe for upper and lower sides of the central line. We estimate the quadrupole parameters from the width of the central line. For the present case of the ^{139}La NMR in $LaTiO_3$, the parameters of $\nu_Q=3.8$ MHz and $\eta \sim 0.6$ are obtained for accounting the width of the central line of $\sim$2.2 kOe. These values are quite reasonable as compared with the results of the zero-field La NMR.

We measured ^{139}La-NMR spectrum up to 200 K. The full width of half amplitude of the central peak is plotted as a function of temperature in Fig. 3 and is almost temperature independent. The NMR shift $K(\sim -0.2\%)$ is also temperature independent in the range of 4.2–220 K as shown in Fig. 4. We did not observe any broadening of the central line below T_N. These results indicate that internal fields through transferred hyperfine interactions from the nearest-neighbor Ti^{3+} moments are almost canceled out at the La site in the antiferromagnetic $LaTiO_3$. We have reported the preliminary results[32] and a similar result was also reported by Itoh, Tanaka, and Motoya.[33] The results give us a hint for understanding of magnetic structure of the antiferromagnetic order of Ti moments in $LaTiO_3$, which will be discussed in the following section.

B. NMR of Sr-substituted $LaTiO_3$

1. NMR of ^{139}La

In the Sr-substituted samples of $La_{1-x}Sr_xTiO_3$, we observed ^{139}La NMR with sweeping extrenal field. ^{139}La NMR shows the quadrupole broadened spectrum similar to the case of $x=0$. The line width of the central peak decreases with increasing x as seen in Fig. 5. The spread of the satellite spectrum decreases also with increasing x. As discussed in the previous section, the main source of the broadening of the central and satellite lines of the La spectrum of $LaTiO_3$ is attributed to the quadrupole effects. Thus, the decrease of linewidth of the central line is considered to be mainly due to the decrease of ν_Q with increasing x. To confirm the scenario that the ΔH is mainly originated from splitting of the central transition ($|1/2\rangle \leftrightarrow |-1/2\rangle$) due to the second-order quadrupole effect, the ^{139}La-NMR spectrum for $x=0.1$ and 0.2 is measured at different resonance frequencies. The observed ΔH is proportional to $1/\nu_0$, as expected from Eq. (4).[29]

A numerical calculation of EFG components is done by the point-charge model with the valence states of La^{3+}, Sr^{2+}, Ti^{3+}, and O^{2+}, and with the lattice constant parameters for orthorhombic symmetry. Summed over the contributions within a sphere of radius $R=20$ A, η is obtained to be 0.65 for $x=0$. When we take the cubic symmetry for $LaTiO_3$, the

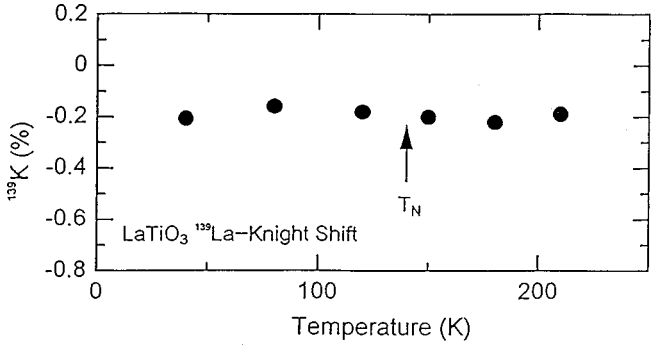


FIG. 4. Knight shift of ^{139}La NMR as a function of temperature for $LaTiO_3$.

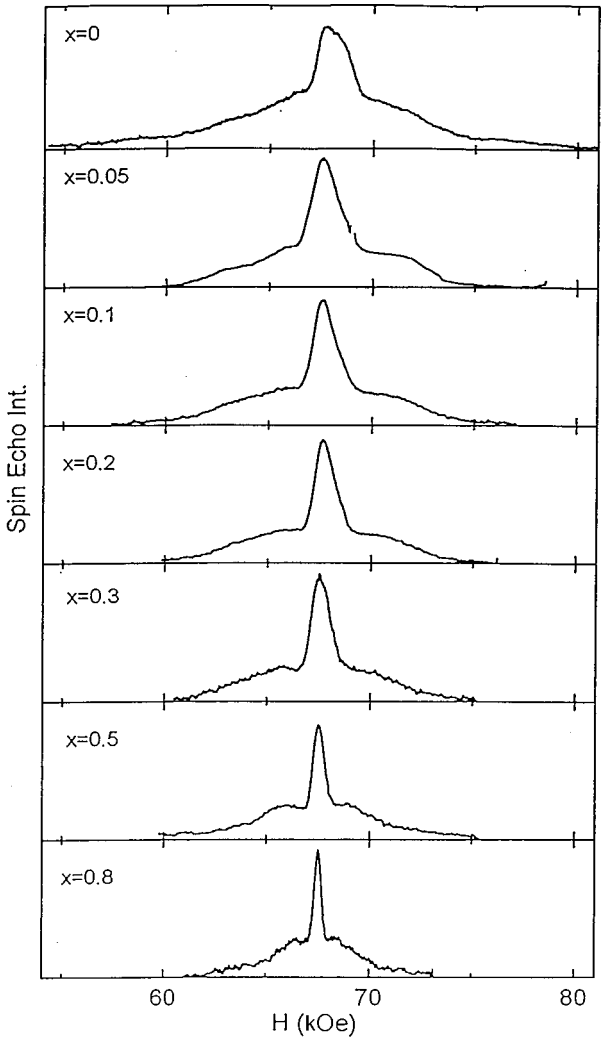


FIG. 5. ^{139}La-NMR spectra as a function of external field for various x in $La_{1-x}Sr_xTiO_3$ obtained at $T=10$ K and $f=40.64$ MHz.

η value does not so much change. Thus, we may simply assume that $\eta\sim0.6$ does not change with x. With this assumption, ν_Q for $La_{1-x}Sr_xTiO_3$ is estimated from ΔH of ^{139}La-NMR spectrum, which are plotted as a function of x in Fig. 6. The accuracy of ν_Q becomes worse for $x>0.3$, where inhomogeneous linewidth may be comparable to the second-order quadrupole splitting.

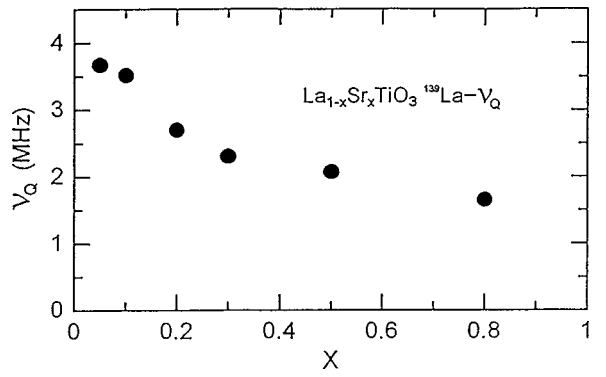


FIG. 6. ν_Q as a function of x for $La_{1-x}Sr_xTiO_3$. Close circles are obtained from NMR results.

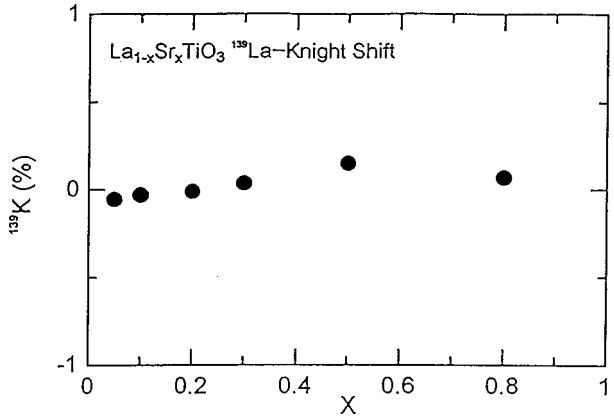


FIG. 7. ^{139}La-Knight shift as a function of temperature in $La_{1-x}Sr_xTiO_3$.

The Knight shifts (^{139}K) defined at peak intensity position of the spectrum is temperature independent for all the Sr-doped samples. (We have taken into account the effects of quadrupole splitting for the Knight shift for $x<0.2$). ^{139}K increases negatively with decreasing x for small x as shown in Fig. 7. This is due to an increases of negative contributions from the spin polarization of La-5d orbitals (so-called core polarization[31]) for ^{139}K. This result suggests that the La-5d state also composes the Fermi level and local density of states at the La site increases with decreasing x, which is shown more clearly in the T_1 results as described later.

Nuclear spin-lattice relaxation time T_1 of ^{139}La is measured by a saturation recovery method. In the case for $I=7/2$, the nuclear magnetization recovery after a saturation pulse is given by,[34]

$$[M(^{\infty})-M(t)]/M(^{\infty})=0.013e^{-2Wt}+0.068e^{-12Wt}+0.206e^{-30Wt}+0.714e^{-56Wt}, \quad (5)$$

where $M(t)$ is the magnetization at a time t after the saturating pulse and $2W=T_1^{-1}$. Experimental recovery curves of ^{139}La nuclear magnetization for the present Sr-substituted samples are well fitted by Eq. (5). The temperature dependence of T_1^{-1} for various x is plotted in Fig. 8. As can be seen, T_1^{-1} for all the samples follow the Korringa relation of

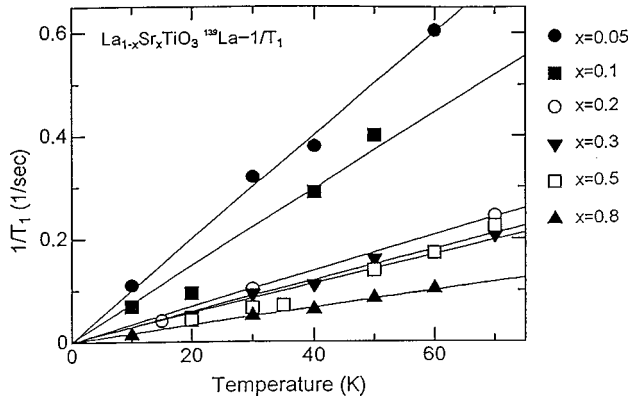


FIG. 8. Temperature dependence of $1/T_1$ of ^{139}La in $La_{1-x}Sr_xTiO_3$. The solid lines represent the $T_1T=$const. relation.

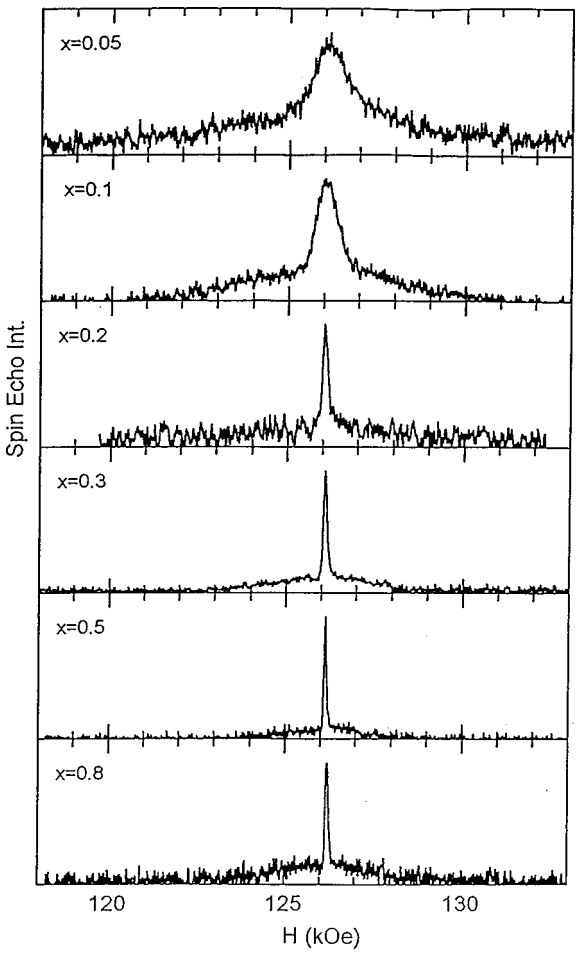


FIG. 9. $^{47/49}$Ti-NMR spectra as a function of external field for various x of $La_{1-x}Sr_xTiO_3$ obtained at $T=10$ K and $f=30.27$ MHz.

$T_1T=$const, indicating that the system is in a metallic state from a microscopic point of view. The $1/T_1$ is enhanced with decreasing x.

2. NMR of $^{47/49}$Ti

$^{47/49}$Ti-NMR experiments have been done using a 15 T superconducting magnet in order to improve signal sensitivity, as ^{47}Ti and ^{49}Ti nuclei have small gyromagnetic ratios of 0.24 005 and 0.2400 (MHz/kOe),[31] and a small natural abundance of 5.5% and 7.5%, respectively. $^{47/49}$Ti-NMR spectra as a function of external field for $La_{1-x}Sr_xTiO_3$ are shown in Fig. 9. Because of the small difference of the nuclear gyromagnetic ratio of the two Ti isotopes, the corresponding peaks for the two isotopes were unresolvable by the measurement at 30.27 MHz. The spectra consist of a relatively sharp central line and much broad satellites. The linewidth of central transition of $^{47/49}$Ti-NMR become sharper with increasing x, as similar to the case of ^{139}La-NMR spectrum, which is indicative of the decrease of quadrupole frequency. For the nondoped $LaTiO_3$, ν_Q of 5.9 (2.3) MHz for ^{47}Ti(^{49}Ti) is already obtained from the zero-field NMR in the antiferromagnetic state. The quadrupole frequency of Ti for various x are estimated from the linewidth of central transition through the procedure similar to the case of La. Although ν_Q values are not accurate for $x>0.3$ where the splitting due to the quardrupole effect may be less than inhomogeneous broadening, we would like to emphasize the increase of ν_Q of Ti as well as La on approaching to $x=0$.

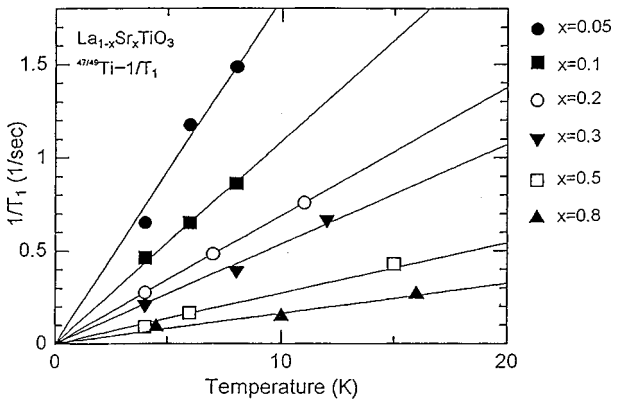


FIG. 10. Temperature dependence of $1/T_1$ of $^{47/49}$Ti in $La_{1-x}Sr_xTiO_3$. The solid lines represent the $T_1T=$const. relation.

T_1 was measured at the peak of the Ti-NMR spectrum by a saturation recovery method. As ^{47}Ti ($I=5/2$) and ^{49}Ti ($I=7/2$)NMR central lines superpose at the peak of the spectrum, we have to take the contributions from the two isotopes for the nuclear magnetization into consideration. Then, we take a following equation for fitting the experimental recovery curves,[35]

$$[M(\infty)-M(t)]/M(\infty)=M_{47}[0.013e^{-2Wt}+0.068e^{-12Wt}+0.206e^{-30Wt}+0.714e^{-56Wt}]+M_{49}[1/35e^{-2Wt}+8/45e^{-12Wt}+50/63e^{-30Wt}]. \quad (6)$$

Here, M_{47} ($=0.655$) and M_{49} ($=1-M_{47}=0.345$) are coefficients for the intensities, which are normalized by natural abundance for the ^{47}Ti and ^{49}Ti isotopes. The experimental data are well fitted by Eq. (6). T_1 is plotted as a function of temperature in Fig. 10. T_1 for the each sample seems to obey the Korringa relation of $T_1T=$const. Figure 11 shows the x dependences of $1/(T_1T)$ of Ti and La estimated from the NMR spectra. The values of $1/(T_1T)$ for the both nuclei increase on approaching to $x=0$.

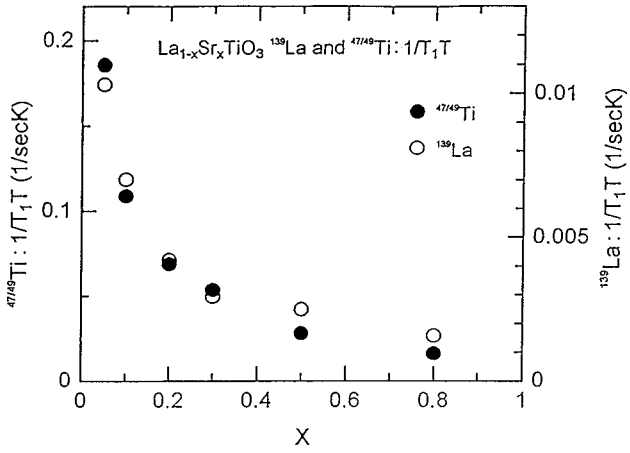


FIG. 11. x dependence of $1/T_1T$ of ^{139}La and $^{47/49}$Ti for $La_{1-x}Sr_xTiO_3$.

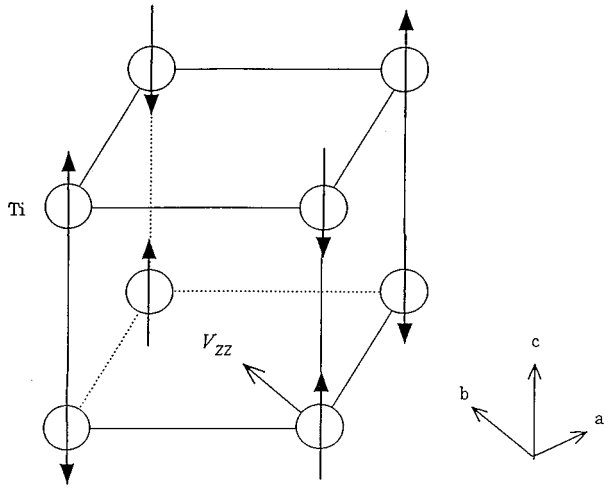


FIG. 12. Schematic view of magnetic structure of $LaTiO_3$.

IV. DISCUSSION

As mentioned in the previous section, the resonance spectrum for Ti in the antiferromagnetically ordered state can be fitted by the set of parameters with the large internal field and electric quadrupole interactions and $\theta \sim 90°$. Here, θ is the angle between magnetic hyperfine field and maximum principal axis of EFG. The maximum principal axis at the Ti and the La site in $LaTiO_3$ is determined to be the b axis by the calculation of EFG by a point-charge model.[30] The NMR shift K ($\sim -0.2\%$) of ^{139}La is temperature independent in the range of 4.2–220 K, and no magnetic broadening of the central line is observed below T_N. These results indicate that internal field through transferred hyperfine interactions from the nearest-neighbor Ti^{3+} moments is almost canceled out at the La site in the antiferromagnetic $LaTiO_3$. This leads to the conclusion that the Ti^{3+} moments are determined to align perpendicular to b axis of the principal axis of EFG as shown in Fig. 12. A slight canting of Ti^{3+} spins ($\theta \sim 1°$) at low temperature reported by magnetization measurement[14] can not distinguish within our experimental uncertainty.

ν_Q of ^{139}La increases as the metal-insulator boundary is approached near $x=0$. This increase can not be explained only by the change of lattice constant with x. One might consider that the large variation of ν_Q with x is to be attributed to a change of crystal structure from orthorhombic to cubic symmetry near $x=0.4$, However, this structural distortion do not significantly affect ν_Q from a point-charge calculation. Alternative interpretation is that the large change of ν_Q is attributed to the change of carrier density at the atomic site as likely to the case for ν_Q in high-T_c cuprates.[36]

Now, we discuss characteristic properties of electronic correlations in $La_{1-x}Sr_xTiO_3$ from microscopic points of view. It is interesting to compare the carrier-doping effects on the evolution of electronic state between the HTSC cuprates with the charge-transfer gap and the $La_{1-x}Sr_xTiO_3$ with the Mott-Hubbard gap. In the case of HTSC, $1/(T_1T)$ of ^{63}Cu in CuO_2 planes increases largely with decreasing temperature,[26–28] but T_1^{-1} or both ^{89}Y and ^{17}O obey the Korringa relation in the normal state.[37] The behavior is explained as follows.

In general, $1/T_1$ is expressed in terms of imaginary part of dynamical spin susceptibility $\chi(q,\omega)$

$$1/T_1 = (\gamma^2 k_B T/2\mu_B^2)\sum A_q \,\mathrm{Im}\,\chi(q,\omega_N)/\omega_N, \qquad (7)$$

where A_q is q-dependent hyperfine-coupling constant and ω_N is the Larmor frequency. A large enhancement of $1/T_1T$ of Cu in HTSC is attributed to the growth of $\chi(Q,\omega_N)$ at antiferromagnetic wave vector Q. For O and Y sites, $\mathrm{Im}\,\chi(Q,\omega_N)$ components do not contribute to T_1^{-1}, because the A_q at O and Y sites vanishes at $q=Q$ due to the geometrical cancellation,[38] so that the $T_1T=$const. behavior is mainly originated from $\mathrm{Im}\,\chi(0,\omega_N)$. Therefore, the large enhancement of $1/T_1T$ at the Cu sites in HTSC is considered to be attributed to the growing of the AF spin correlations with decreasing temperature.[39] Such AF spin correlations are enhanced progressively toward to the AF insulator phase from the metallic side.[40]

Keeping this in mind, we discuss the T_1 results for both Ti and La in $La_{1-x}Sr_xTiO_3$. Antiferromagnetic-spin fluctuations between spins on the Ti^{3+} ($S=\frac{1}{2}$) ions are canceled out at the La sites, because the La atoms located in the symmetrical position between the Ti sites. This can be easily confirmed from the fact that the internal field at the La sites though the transferred hyperfine interaction from Ti^{3+} moments is close to zero as described already. Therefore, we may consider that the spin fluctuations with the $q \sim Q$ component is almost filtered out at the La site due to the geometrical cancellations, as similar to the case of the oxygen sites in CuO_2 plane or the yttrium sites in HTSC.[37,38] Thus, the La nuclei in $La_{1-x}Sr_xTiO_3$ experience only the spin fluctuations at $q \sim 0$ component. In contrast, the Ti nuclei are expected to be affected by the spin fluctuations with all q components. We consider the case that $1/T_1T$ is simply expressed by

$$1/T_1T = (\gamma^2 k_B/2\mu_B^2)A_0\,\mathrm{Im}\,\chi(0,\omega_N)/\omega_N \quad \text{for La,}$$

and

$$1/T_1T = (\gamma^2 k_B/2\mu_B^2)\sum A_q\,\mathrm{Im}\,\chi(q,\omega_N)/\omega_N \quad \text{for Ti.}$$

As T_1 of both Ti and La at low temperature follows the Korringa relation of $T_1T=$const, we consider there are no significant growth of antiferromagnetic spin correlations with decreasing temperature. The experimental results show also that $1/T_1T$ of Ti and La change similarly with x as shown in Fig. 11. These lead to the conclusion that $\mathrm{Im}\,\chi(q,\omega_N)$ changes in similar way at all q components as the Mott transition is approached from the metallic side, and that there is no progressive growth of the $\mathrm{Im}\,\chi(q=Q,\omega_N)$ components due to antiferromagnetic-spin correlations with decreasing x. Thus, the AF spin fluctuations of Ti-3d spins due to the interatomic electronic correlations is not important, but *single site* (*intraband*) *electronic correlations play an important role in driving the Mott insulator-metal transition*, as opposed to the HTSC case where the interatomic AF spin fluctuations of Cu spins are enhanced as toward to the metal-insulator boundary.

In order to discuss qualitatively magnetic correlations in metals, it is useful to see the Korringa relation in detail. The modified Korringa relation from a free-electron model is given by,[41,42]

$$T_1TK^2_{\rm spin}=S/K(\alpha),$$

where S is a so-called Korringa value given by $(\gamma_c^2/\gamma_N^2)(h/4\pi k_B)$ and a factor, $K(\alpha)$ reflects the electron correlations. The factor $K(\alpha)$ obtained for a random-phase approximation (RPA) is given by $K(\alpha)=\langle(1-\alpha)^2/\{1-\alpha[\chi_0(q)/\chi_0\}^2]\rangle_{FS}$. Here, $\chi_0(q)$ is a dynamical susceptibility for a noninteracting system and χ_0 is its uniform component. $\langle\rangle_{FS}$ denotes the average over wave vectors on the Fermi surface and $1/(1-\alpha)$ is the Stoner enhancement factor. Taking the intraatomic Coulomb interactions into account in a RPA framework, Moriya,[41] and Narath and Weaver[42] show that $K(\alpha)$ is close to unity when $\chi(q)$ is almost uniformly enhanced in the q space, while $K(\alpha)$ is very small, of the order of $(1-\alpha)$, in the case that $\chi(q)$ is enhanced only in a small-q region. Therefore, $K(\alpha)$ is expected to be larger than 1 for the case of $\chi(q)$ enhanced at the finite wave vector of $q=Q$ in $La_{1-x}Sr_xTiO_3$.

In order to estimate $K(\alpha)$ from the experimental results, we need the value of $K_{\rm spin}$ which is usually evaluated from a $K-\chi$ plot. In the present case, however, it is difficult to estimate accurate values of $K_{\rm spin}$ for Ti, because the temperature dependence of Knight shift is so small to determine accurate H_{hf} value, which may be partly due to the smallness of the nuclear magnetic moments. Therefore, we estimate values of $K(\alpha)$ using the experimental results of magnetic susceptibility and specific heat.[5,19] $\chi_0(0)$ is estimated from the value of γ by using the relation, $\chi_0(0)=6\gamma\mu_B^2/\pi^2k_B^2$. From the experimental values for χ,$1/(1-\alpha)$ are estimated to be nearly 2 leading to $\alpha\sim0.5$ for all x. Thus, $K(0.5)$ is estimated to be 0.72, which is not far from unity. Therefore, we may conclude that $\chi(q)$ is enhanced almost uniformly over the whole q space.

The characteristics of Mott transition in the Ti oxide is turned out to be different from the Cu oxide. In the case of HTSC, antiferromagnetic long range order is destroyed by spin frustration due to excess doped hole s spins ($S=\frac{1}{2}$) on the p orbitals of oxygen, resulting in large AF spin fluctuations between Cu spins, For the Ti oxides, on the other hand, the doped holes occupies not O-2p orbitals but mainly Ti-3d orbitals. No obvious growth of AF spin correlations in the Ti oxides can be attributed to the importance of the single-site (intraband) electronic correlations for driving the Mott insulator-metal transition.

Finally, we consider density of states or effective mass of the conduction electrons on the Fermi surface at Ti and La sites utilizing the $1/T_1T$ results, which follows the Korringa relation. Since $1/T_1$ is enhanced due to electronic correlations, we simply assumed that the enhancement factor $\langle(1/\{1-\alpha(\chi_0(q)/\chi_0)\}^2\rangle_{FS}$, is 2.88 with $K(0.5)=0.72$ and $\alpha=0.5$. We also assume that this is independent of x. $1/T_1T$ in a metallic state is decomposed as,[43]

$$1/T_1T=(1/T_1T)_s+(1/T_1T)_{\rm orb}+(1/T_1T)_{\rm cp}+(1/T_1T)_{\rm dipole},$$

where $(1/T_1T)_s$, $(1/T_1T)_{\rm orb}$, $(1/T_1T)_{\rm cp}$, and $(1/T_1T)_{\rm dipole}$ represent the Fermi contact (for s electron), d orbital, d core polarization, and d dipole contributions, respectively. In the present case, $^{\rm Ti}(1/T_1T)$ for Ti can be attributed to the contribution from the d orbital and the d core polarization, since the $(1/T_1T)_s$ and $(1/T_1T)_{\rm dipole}$ terms can be neglected. Here

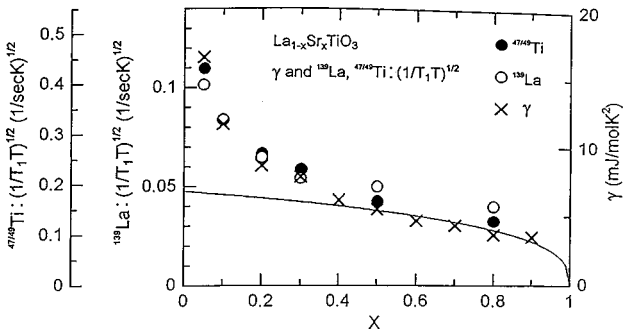


FIG. 13. x dependences of $1/\sqrt{T_1T}$ and the electronic specific heat coefficient, γ (from Ref. 19) of $La_{1-x}Sr_xTiO_3$.

$(1/T_1T)_i=4\pi\gamma_e\gamma_nhA_i^2N(E_F)^2F_i$ ($i=$orb and cp), where A_i are hyperfine-coupling constants for i, reduction factors F_i arising from orbital degeneracy is 0.6. $N(E_F)$ is the renormalized density of states at the Fermi level [The enhancement effects are renormalized in $N(E_F)$]. As the La-5d bands exist at the Fermi level, the $^{\rm La}(1/T_1T)$ is also decomposed as $^{\rm La}(1/T_1T)=^{\rm La}(1/T_1T)_{\rm orb}+^{\rm La}(1/T_1T)_{\rm cp}$.

Assuming $A_{\rm cp}=-100\,{\rm kOe}/\mu_B$, $A_{\rm orb}=160\,{\rm kOe}/\mu_B$, $F_{\rm cp}=0.4$, and $F_{\rm orb}=0.2$ for Ti, and $A_{\rm cp}=-1200\,{\rm kOe}/\mu_B$, $A_{\rm orb}=180\,{\rm kOe}/\mu_B$, $F_{\rm cp}=0.4$, and $F_{\rm orb}=0.2$ for La[35,42] we estimate $N(E_F)$ of the La and the Ti site. Figure 13 shows the x dependence of $1/\sqrt{T_1T}$ [$\propto N(E_F)$] of La and Ti together with the result of electronic specific heat coeffient, $\gamma(\propto {}^{\rm total}N(E_F))$. In spite of our crude estimations of $N(E_F)$, $^{\rm Ti}N(E_F)$ of the Ti site are in good agreement with $^{\rm total}N(E_F)$. Figure 14 shows x dependence of the ratio of $^{\rm Ti}N(E_F)$ to $^{\rm La}N(E_F)$ ($\equiv R$). The nearly x independent R ($\sim$45) shows that $N(E_F)$ at the Ti site is 45 times larger than that at the La site for all $La_{1-x}Sr_xTiO_3$. These results reveal microscopically that doped holes occupy mainly Ti-3d bands.

Here, we would like to stress that the critical enhancement of $N(E_F)$ is apparently demonstrated near the phase boundary of the Mott transition.[19] In a parabolic band for noninteracting electrons, $N(E_F)$ is given by $N(E_F)=(4\pi/h^3)(2m^*)^{3/2}E_F^{1/2}$, where $E_F=(h^2/2m^*)(3\pi^2n_0)^{2/3}$. Here, n_0 is the carrier density and m^* is renormalized effective electron mass. Therefore, we get a simple relation of $(T_1T)^{-1/2}\propto(m^*n_0^{1/3})$. for a noncorrelated or a rigid band

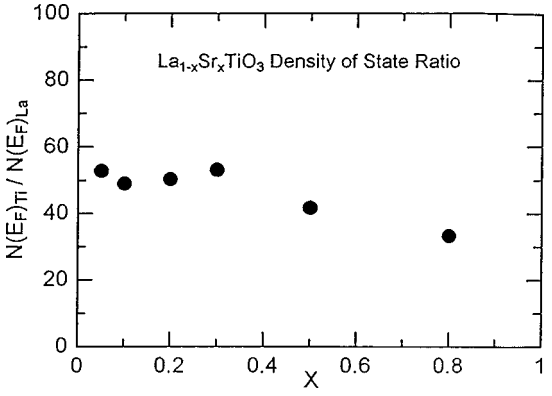


FIG. 14. Ratio of the density of states of the Ti and the La site, $^{\rm Ti}N(E_F)/^{\rm La}N(E_F)$ as a function of x.

model. From measurements of Hall coefficients in $La_{1-x}Sr_xTiO_3$,[5,19] n_0 is found to increase linearly with decreasing x. Therefore, the deviation of $(T_1T)^{-1/2}$ for both Ti and La near $x\sim 0$ from the simple relation of $(T_1T)^{-1/2} \propto (m^* n_0^{1/3})$ provide microscopically the critical enhancement of the effective carrier mass near the phase boundary of the Mott transition in the system, as demonstrated previously by the specific heat measurement.[5,18,19]

V. CONCLUSION

We have measured $^{47/49}$Ti and ^{139}La NMR to investigate carrier-doping effects on electronic correlations in $La_{1-x}Sr_xTiO_3$. In antiferromagnetic insulator $LaTiO_3$ with $T_N\sim 140$ K, $^{47/49}$Ti-NMR spectra under zero external field provides the internal magnetic field of 91.4 kOe at Ti. On the other hand, the hyperfine field at the La sites is canceled out at the AF ordered state. Together with the results of the quardrupole interactions, we obtain the magnetic structure of the Ti moments with $S=\frac{1}{2}$ in the AF state as shown in Fig. 12.

In the Sr-substituted samples, T_1^{-1} of both La and Ti obey the Korringa relation of $T_1T=$const., indicative of the absence of the growth of AF spin correlations at $q\sim Q$ with decreasing temperature. From the x dependences of $(T_1T)^{-1}$ for both La and Ti, the q-dependent dynamical spin susceptibility is confirmed to change in a similar way at all q components with x. We have no evidences for a prominent increase of antiferromagnetic spin fluctuations at $q\sim Q$ on approaching the Mott transition from the metallic side. Thus, we conclude that the AF spin fluctuations of Ti-3d spins due to the *interatomic* electronic correlations is not important, but *single-site* (*intraband*) electronic correlations play an important role in driving the Mott insulator-metal transition, as opposed to the HTSC case. The density of state at Fermi level of Ti sites is about 45 times larger than that of La sites, suggesting Ti3d-band is main part at the Fermi level in $La_{1-x}Sr_xTiO_3$.

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